🎲 Lesson: Introduction to Probability

1. What is Probability?

Probability measures how likely an event is to occur. It is a number between **0** and **1**: - **0** means the event will *never* happen. - **1** means the event will *certainly* happen.

For any event **A**:

[P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}]

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2. Basic Terminology

Term	Definition	Example
Experiment	A process that leads to an outcome	Rolling a die
Sample Space (S)	The set of all possible outcomes	{1, 2, 3, 4, 5, 6}
Event (A)	A subset of the sample space	{2, 4, 6} → even numbers
Outcome		Rolling a 4

Term	Definition	Example
	A single result from the experiment	

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3. Types of Probability

(a) Theoretical Probability

Calculated by reasoning or symmetry: [P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}]

Example: Probability of getting a 3 when rolling a fair die: [P(3) = \frac{1}{6}]

(b) Experimental Probability

Based on observation or experiment: [P(A) = \frac{\text{Number of times A occurred}}{\text{Total number of trials}}]

(c) Subjective Probability

Based on intuition, belief, or experience.

Example: Probability that it rains tomorrow (based on weather forecast).

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4. Complementary Events

For any event **A**, the probability that **A does not occur** is: [P(A') = 1 - P(A)]

Example: If (P(A) = 0.3), then (P(A') = 1 - 0.3 = 0.7).

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5. Independent and Dependent Events

Type	Definition	Example
Independent	One event does not affect the other	Rolling two dice
Dependent	One event affects the other	Drawing two cards without replacement

For independent events **A** and **B**: [P(A \text{ and } B) = P(A) \times P(B)]

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6. Conditional Probability

The probability of **A** given that **B** has already occurred: [P(A|B) = \frac{P(A \cap B)}{P(B)}]

Example: If 40% of students play sports, and 25% both play sports and play chess, then: [P(\text{Chess | Sports}) = \frac{0.25}{0.40} = 0.625]

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7. Addition Rule of Probability

For any two events **A** and **B**: [P(A \cup B) = P(A) + P(B) - P(A \cap B)]

If **A** and **B** are mutually exclusive (cannot happen together): [P(A \cup B) = P(A) + P(B)]

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8. Law of Total Probability

If events (A_1, A_2, \ldots, A_n) are mutually exclusive and exhaustive: [P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \ldots + P(A_n)P(B|A_n)]

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9. Bayes' Theorem

Used to update probabilities based on new information: [P(A|B) = \frac{P(B|A)P(A)}{P(B)}]

Example: In medical testing, it helps find the probability a patient actually has a disease given a positive test.

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10. Random Variables

A **random variable** assigns a numerical value to outcomes of a random process. - **Discrete:** Finite number of outcomes (e.g., dice rolls) - **Continuous:** Infinite outcomes within an interval (e.g., height, time)

Expected Value (Mean):

[E(X) = \sum x_i P(x_i)]

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11. Common Probability Distributions

Distribution	Type	Example
Bernoulli	Discrete	Success/Failure
Binomial	Discrete	Number of successes in n trials
Poisson	Discrete	Number of events in fixed time
Uniform	Continuous	Equal likelihood in a range
Normal (Gaussian)	Continuous	Heights, test scores

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12. Practice Problems

1. A fair coin is tossed twice. Find the probability of getting:
2. (a) Two heads
3. (b) At least one head
4. A bag contains 3 red, 4 blue, and 5 green balls. One ball is picked at random. Find the probability that it is:
5. (a) Red
6. (b) Not blue
7. The probability that it rains on Monday is 0.6 and on Tuesday is 0.5. Assuming independence, find the probability it rains on both days.

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13. Summary Table

Concept	Formula
Complement	(P(A') = 1 - P(A))
Intersection (AND)	(P(A \cap B) = P(A)P(B)) (if independent)
Union (OR)	(P(A \cup B) = P(A) + P(B) - P(A \cap B))
Conditional	(P(A
Bayes' Theorem	(P(A

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14. References

- Sheldon Ross, *A First Course in Probability*
- Blitzstein & Hwang, *Introduction to Probability*
- Khan Academy: https://www.khanacademy.org/math/statistics-probability

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> ***Tip:*** *Always start by identifying the sample space and defining the event clearly before applying any formula.*

Generated on 2025-11-11 22:57:13